

Assignment 5 & 6, 2025

Level: BE

Subject : Algebra and Geometry

Due Date : 31 July, 2025

Attempt all questions.

1. Transform the equation $x^2 + 3y^2 + 3x - 40 = 0$ with origin through the point (4,-1).
2. Find angle of rotation of axes so that the equation $ax^2 + 2hxy + by^2$ has no product term xy .
3. What does the equation of pair of straight lines $7x^2 + 4xy + 4y^2 = 0$ becomes when the axes are the bisectors of angle between them?

Hint:

1. Compare the given equation with $ax^2 + 2hxy + by^2 = 0$
2. Then the equations of angle of bisectors of given pair of straight lines are given by $h(x^2 - y^2) = (a - b)xy$.
3. Then, we get two pair of straight lines $(2x + y) = 0$ and $(x - 2y) = 0$
4. Slope of one bisector is $\tan\theta = \frac{1}{2}$ where θ is the angle between bisector line and x-axis.
5. Then, we have $\sin\theta = \frac{1}{\sqrt{5}}$ and $\cos\theta = \frac{2}{\sqrt{5}}$.
6. Transform the given equation by rotating the axis by θ . That is, in the original equation, replace x and y by $x = \frac{2X-Y}{\sqrt{5}}$ and $y = \frac{X+2Y}{\sqrt{5}}$, then we get the require equation.
4. Find the centre, vertex and eccentricity, foci, length of major and minor axes, length of latus rectum of the ellipse $9x^2 + 4y^2 + 40y + 18x + 73 = 0$.
5. Find the equation of an ellipse having focus at (-1,1), eccentricity (e)=1/2 and directrix: $x-y+3 = 0$.
6. Find the equation of tangent and normal to the ellipse $3x^2 + 4y^2 = 84$ at point (4,3).
7. Find the equation of tangent to the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$ which is parallel to the line $x = y + 4$.
8. Show that the line $3x + 4y + \sqrt{7} = 0$ touch the ellipse $3x^2 + 4y^2 = 1$. Also, find the point of contact.
9. Find the value of c so that the line $2x - 3y = c$ is normal to the ellipse to the ellipse $9x^2 + 116y^2 = 144$.
10. Find the centre, eccentricity, foci and the directrix of the hyperbola $9x^2 - 16y^2 - 72x + 96y - 144 = 0$.
11. Find the equation of the tangents to the hyperbola $3x^2 - 4y^2 = 12$, which are perpendicular to the line $y = x + 2$. Also, find the point of contact.
[Hint: Example 29, page 207].
12. Show that the equation $16x^2 - 24xy + 9y^2 - 104x - 172y + 44 = 0$ represents a parabola. Also, determine its equation of axis, vertex, equation of Latus rectum, length of Latus rectum, focus and equation of directrix.
13. Determine the nature of the conic section represented by $14x^2 - 4xy + 11y^2 - 44x - 58y + 71 = 0$. Also, find the centre, length of semi axes, length of latus rectum, equation of axes, eccentricity, foci, directrices of the conic section.

14. Determine the nature of the conic section represented by $x^2 + 12xy - 4y^2 - 6x + 4y + 9 = 0$. Also, find the centre, length of semi axes, length of latus rectum, equation of axes, eccentricity, foci, directrices of the conic section.
15. Sketch and describe the conic section $r = \frac{6}{1 - \cos\theta}$.
16. Sketch and describe the conic section $r = \frac{12}{3 + 2\cos\theta}$.
17. Sketch and describe the conic section $r = \frac{12}{2 - 3\sin\theta}$.
18. Find the distance of the point (3,-4,5) from the plane $2x + 5y - 6z = 16$ measured along a line with direction cosines proportional to 2,1,-2. (That is direction ratios are 2,1,-2)
[Hint: Example 2, page 230].
19. Reduce the line $x + 2y + 3z - 6 = 0 = 3x + 4y + 5z - 2$ in the symmetrical form.
20. Find the point where the line joining (1,2,3) and (3, -1, 4) cuts the plane $3x + 2y + z - 2 = 0$.
21. Find the two points on the line $\frac{x-1}{1} = \frac{y+3}{2} = \frac{z+5}{2}$ on either side of (2,-3,-5) and at a distance 3 from it.
22. Find the image of the point P(1,3,4) in the plane $2x - y + z + 3 = 0$.
[Hint: Example 3, page 231].
23. Prove that the lines $x = ay + b$, $z = cy + d$ and $x = a'y + b'$, $z = c'y + d'$ are perpendicular if $aa' + cc' + 1 = 0$.
24. Show that the lines $\frac{x+1}{-3} = \frac{y-3}{2} = \frac{z+2}{1}$ and $x = \frac{y-7}{-3} = \frac{z+7}{2}$ are coplanar. Also, find the equation of the plane containing them.
25. Prove that the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $4x - 3y + 1 = 0 = 5x - 3z + 2$ are coplanar. also, find the equation plane containing them.
26. Find the shortest distance between two skewed lines $\frac{x}{2} = \frac{y}{-3} = \frac{z}{1}$ and $\frac{x-2}{3} = \frac{1-y}{5} = \frac{z+2}{2}$. Also, find the equation of line of shortest distance.
27. Find the length of perpendicular between the skewed line, $x + a = 2y = -12z$ and $x = y + 2a = 6z - 6a$.
28. Find the equation of the sphere which passes through the points (1,-4,3), (1,-5,2) and (1,-3,0) and whose centre lies on the plane $x + y + z = 0$.
29. Obtain the equation of the sphere having the circle $x^2 + y^2 + z^2 + 10y - 4z - 8 = 0$, $x + y + z = 3$ as a great circle.
30. Find the equations of the sphere through the circle $x^2 + y^2 + z^2 = 1$, $6x - 8y + 2z + 9 = 0$ and touching the plane $y = 0$.
[Hint: Example 19, page 257].
31. A sphere of radius k passes through the origin and meets the axes in A, B, C. Prove that the centroid of the triangle ABC lies on the sphere $9(x^2 + y^2 + z^2) = 4k^2$.
[Hint: Example 16, page 254].

32. A sphere S has points $(0,1,0)$, $(3,-5,2)$ as opposite ends of a diameter. Find the equation of the sphere having the intersection of the sphere S with the plane $5x - 2y + 4z + 7 = 0$ as a great circle.
33. Show that the equation of the sphere through the circle $x^2 + y^2 + x^2 - 2x - 3y + 4z + 8 = 0$, $x^2 + y^2 + z^2 + 4x + 5y - 6z + 2 = 0$ and having its centre on the plane $4x - 5y - z = 3$ is $x^2 + y^2 + z^2 + 7x + 9y - 11z - 1 = 0$.
34. Show that the plane $2x - y + 2z = 14$ touches the sphere $x^2 + y^2 + z^2 - 4x + 2y - 4 = 0$. Also, find the point of contact.
35. Find the equation of the tangent plane to the sphere $x^2 + y^2 + z^2 = 49$ at the point $(6,-3,2)$.
36. Find the equation of the sphere having its centre $(3,4,5)$ and touching the line $\frac{x-1}{1} = \frac{y+2}{2} = \frac{z+2}{-2}$.
[Hint: Example 25, page 264].
37. Find the equations of the tangents planes to the sphere $x^2 + y^2 + z^2 + 6x - 2z + 1 = 0$ which passes through $\frac{16-x}{1} = z = \frac{2y+30}{3}$.
[Hint: Example 26, page 265].
38. Find the equations of the tangents planes to the sphere $x^2 + y^2 + z^2 - 2x + 4y - 6z + 10 = 0$ which passes through $\frac{x+3}{14} = \frac{y+1}{-3} = \frac{z-5}{4}$.
Hint: Taking first two ratios, we get, $3x + 14y + 23 = 0$ and taking last two ratios, we get, $4y + 3z - 11 = 0$ then equation of line in general form is $3x + 4y + 23 = 0 = 4y + 3z - 11$. Thus, consider the equation of plane passing through the given line as
- $$4y + 3z - 11 + \lambda(3x + 4y + 23) = 0$$
39. Find the equation of a cone which has vertex at origin and passes through the curve $ax^2 + by^2 = 2z$, $lx + my + nz = p$.
40. Find the equation of cone whose vertex is (α, β, γ) and base (guiding curve) is $y^2 = 4ax$, $z = 0$.
41. Find the equation of cone whose vertex is $(1, 1, 0)$ and base (guiding curve) is $x^2 + z^2 = 4$, $y = 0$.
42. Show that the equation to a right circular cone whose vertex is O, axis OX and semi-vertical angle α is $y^2 + z^2 = x^2 \tan^2 \alpha$.
[Hint: Example 33, page 272].
43. Find the equation of a right circular cone with vertex $(1,1,1)$, axis as $\frac{x-1}{-1} = \frac{y-1}{2} = \frac{z-1}{3}$ and semi-vertical angle 30° .
44. Find the equation of cylinder whose generators are parallel to the line $\frac{x}{1} = \frac{y}{-2} = \frac{z}{3}$ and guiding curve is the ellipse $x^2 + 2y^2 = 1$, $z = 0$.
45. Find the equation of right circular cylinder of radius 2 whose axis is the line $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-3}{2}$.
46. Find the equation of right circular cylinder whose guiding curve is $x^2 + y^2 + z^2 - x - y - z = 0$, , $x + y + z = 1$.
[Hint: Example 39, page 277].

47. Find the equation of cylinder which passes through the curve of intersection of plane $lx+my+nz = p$ and $ax^2 + by^2 + cz^2 = 1$ and generators parallel to z-axis.
[Hint: Example 35, page 275].

"ALL THE BEST."